

The empirical recurrence for OEIS A250923: recovering a conjecture that is not written down

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Abstract

OEIS A250923 records “Empirical recurrence of order 81”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 243$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 81, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 81 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A250923 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with no 2×2 subblock having its minimum diagonal element less than its minimum antidiagonal element.”. It has offset 1 and begins

11693, 628146, 34521887, 1887760064, 102912852059, 5613563039740, 306543548743117, 16748448512897363, ...

The entry states:

Empirical recurrence of order 81 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:51:27, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 243$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 243$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 486$ exact terms of the model returns order 81 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{81} c_i a(n-i),$$

c_1	243	c_{22}	−3918416366882006468835718789	c_{43}	3279023424367421
c_2	−27764	c_{23}	19478365433409142384538896162	c_{44}	−462011922966091
c_3	2015179	c_{24}	−90284011239187282069351733917	c_{45}	6156931635469487
c_4	−105182400	c_{25}	390888455515223879548815907826	c_{46}	−775957576920067
c_5	4223631481	c_{26}	−1583323368667679577720656519628	c_{47}	9247187361979527
c_6	−136125966767	c_{27}	6008749364932742160277983941660	c_{48}	−1041812102486016
c_7	3625550001219	c_{28}	−21392307623796565355437493282297	c_{49}	1109329033690686
c_8	−81527377542987	c_{29}	71531243894551728326444205471570	c_{50}	−1116033578081482
c_9	1573726427167963	c_{30}	−224881102511757690091397563153723	c_{51}	1060386547026846
c_{10}	−26425068602011767	c_{31}	665329324911933940382832789021765	c_{52}	−951070290419873
c_{11}	390213315069594779	c_{32}	−1854005153529421086701144817758955	c_{53}	8047824858823601
c_{12}	−5113943537364581235	c_{33}	4869669463688260661839163168623886	c_{54}	−642066038666793
c_{13}	59944541549153011205	c_{34}	−12063918753624964409281159783666566	c_{55}	4826057220284255
c_{14}	−632673010991315794758	c_{35}	28205163465580807780455221872699617	c_{56}	−341466438648924
c_{15}	6047188155526054696614	c_{36}	−62264152489109290784038024191492146	c_{57}	2272105058903993
c_{16}	−52608994250640381463034	c_{37}	129838571823276834639964238361679155	c_{58}	−142024034879011
c_{17}	41842522277267310808028	c_{38}	−255848745954510960156230405911023358	c_{59}	832947080028941
c_{18}	−3054347056740093826010599	c_{39}	476548218130989367040884005455974807	c_{60}	−457718777672845
c_{19}	20533289137307400145143279	c_{40}	−839224584991235137798788547605179349	c_{61}	235307893368606
c_{20}	−127516996856916234046715620	c_{41}	1397577740161730196744985189615144823	c_{62}	−112974219940318
c_{21}	733554306335506463055321894	c_{42}	−2201174794086024696170920188905677437	c_{63}	505571575746110

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 81, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $81 + 4$ terms removed returns order 81 as well, so $\deg q_{\min} = 81 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $81 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 81 that this sequence satisfies — and that family turns out to have exactly one member.

References

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